Exercise 13.3 SOLUTION

Q.1. A car's position at time t is given by $s(t) = 5t^3 - 3t^2 + t$. Find the velocity by differentiating the position function with respect to time.

Sol: $s(t) = 5t^3 - 3t^2 + t$

Take derivative w.r.t. t

$v(t) = s'(t) = 5(3t^2) - 3(2t) + 1 = 15t^2 - 6t + 1$

The velocity function is $v(t) = 15t^2 - 6t + 1$

Q.2. Structural stress on a bridge is modeled by the function $S(x) = 100 - 5x^2$, where x is the distance from the center of the bridge. Calculate the rate of change of stress at that point.

Sol: Stress Function: $S(x) = 100 - 5x^2$

Rate of change of stress is the derivative of $S(x)$ w.r.t. "x".

$$S'(x) = \frac{ds}{dx} = \frac{d}{dx}(100 - 5x^2)$$

$$= \frac{d}{dx}(100) - 5\frac{d}{dx}(x^2)$$

$$= 0 - 5(2x) = -10x$$

Since the center of bridge is at $x = 0$, therefore rate of change at the point is

$S'(0) = -10(0) = 0$

Maximum stress= $S(0) = 100 - 5(0)^2 = 100$

Q.3. A Company's revenue function is given by $R(x) = 1000x - 10x^2$, where is the number of units produced. The cost function is $C(x) = 300x + 2000$.

i. Find the profit function $P(x)$ ii. Determine the marginal profit when $x = 15$

Sol: Calculate the profit function $P(x)$

$P(x) = R(x) - C(x)$

Substitute the given revenue and cost functions:

$P(x) = (1000x - 10x^2) - (300x + 2000)$

Simplify the expression $P(x) = 1000x - 10x^2 - 300x - 2000$

$P(x) = -10x^2 + 700x - 2000$

Determine the marginal profit function $P'(x)$

The marginal profit function is the derivative of the profit function:

$$P'(x) = \frac{d}{dx}(-10x^2 + 700x - 2000)$$

Apply the power rule: $P'(x) = -20x + 700$

Calculate the marginal profit at $x = 15$

Substitute $x = 15$ into the marginal profit function:

$P'(15) = -20(15) + 700 = -300 + 700 = 400$

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Q.4. An investment grows according to the function $A(t) = 10000(1+0.05t)^3$, where $A(t)$ is the value of the investment and t is the time in years.

i. Find the rate of change of the investment after 8 years. ii. What is the investment value after 8 years?

Sol: a) To find the rate of change, we need to find the derivative of the function $A(t)$ with respect to time t.

$$A(t) = 10000(1+0.05t)^3$$

Take derivative w.r.t. t

$$\frac{dA}{dt} = 10000.3(1+0.05t)^{3-1}.\frac{d}{dt}(1+0.05t)$$

$$\frac{dA}{dt} = 30000(1+0.05t)^2.(0.05) = 1500(1+0.05t)^2$$

Now, to find the rate of change after 8 years, we substitute $t = 8$ into the derivative:

$$\frac{dA}{dt}|_{t=8} = 1500(1+(0.05)(8))^2 = 1500(1+0.4)^2$$

$$\frac{dA}{dt}|_{t=8} = 1500(1.4)^2 = 1500(1.96) = 2940$$

So, the rate of change of the investment after 8 years is 2940 per year.

b) **What is the investment value after 8 years?**

Sol: To find the investment value after 8 years, we substitute $t = 8$ into the original function $A(t)$:

$$A(8) = 10000(1+0.05.8)^3$$

$$A(8) = 10000(1+0.4)^3 = 10000(1.4)^3 = 10000(2.744) = 27440$$

Therefore, the investment value after 8 years is 27440.

Q.5. **The position of a particle moving along a line is given by $S(t) = 5t^3 - 12t^2 + 8t$ where $S(t)$ is the position in meters and t is the time in seconds.**

i. Determine the velocity of the particle at $t = 4$ seconds

ii. Find the acceleration at $t = 4$ seconds **iii. When is the particle at rest?**

a) **Velocity at $t = 4$ seconds**

Sol: 1. Position function: $s(t) = 5t^3 - 12t^2 + 8t$

2. Velocity function (first derivative):

$$v(t) = \frac{ds}{dt} = 15t^2 - 24t + 8$$

3. at $t = 4$:

$$v(4) = 15(4)^2 - 24(4) + 8$$

$$v(4) = 15(16) - 96 + 8 = 240 - 96 + 8 = 152 \ \ m/s$$

b) **Acceleration at $t = 4$ seconds:**

Sol: 1. Velocity function: $v(t) = 15t^2 - 24t + 8$

2. Acceleration function (first derivative of velocity, or second derivative of position):

$$a(t) = \frac{dv}{dt} = 30t - 24$$

3. Evaluate at $t = 4$:

$$a(4) = 30(4) - 24$$

$$a(4) = 120 - 24$$

$$a(4) = 96 \ \ m/s^2$$

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c) **When is the particle at rest?**

Sol: $v(t) = 15t^2 - 24t + 8 = 0$

When particle is at rest put $v(t) = 0$

$\Rightarrow 15t^2 - 24t + 8 = 0$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where $a = 15, b = -24, c = 8$

$$t = \frac{-(-24) \pm \sqrt{(-24)^2 - 4(15)(8)}}{2(15)} = \frac{24 \pm \sqrt{576 - 480}}{30} = \frac{24 \pm \sqrt{96}}{30}$$

$$t = \frac{24 \pm \sqrt{96}}{30} = \frac{24 \pm 4\sqrt{6}}{30} = \frac{12 \pm 2\sqrt{6}}{15}$$

The particle is at rest at $t = \frac{12 + 2\sqrt{6}}{15}$ seconds and $t = \frac{12 - 2\sqrt{6}}{15}$ seconds.

Q.6. **The Position of a car traveling along a straight highway is given by $x(t) = 30t^2 - 4t^3$, where $x(t)$ is the distance traveled in kilometers and t is the time in hours.**

a. Find the car's velocity at $t = 3$ hours. b. Determine the car's acceleration at $t = 3$ hours.

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a) **Velocity at $t = 3$ hours.**

Sol: 1. Position function: $x(t) = 30t^2 - 4t^3$

2. Velocity function (first derivative):

$$v(t) = \frac{dx}{dt} = 60t - 12t^2$$

3. Evaluate at $t = 3$:

$$v(3) = 60(3) - 12(3)^2 = 180 - 108 = 72 \quad km/h$$

b) **Acceleration at $t = 3$ hours.**

Sol: 1. Velocity function $v(t) = 60t - 12t^2$

2. Acceleration function (first derivative of velocity):

$$a(t) = \frac{dv}{dt} = 60 - 24t$$

3. Evaluate at $t = 3$:

$$a(3) = 60 - 24(3)$$

$$a(3) = 60 - 72 = -12 \quad km/h^2$$

Q.7. **The stress on a beam under a varying load is given by $S(x) = 400x - x^3$ where $S(x)$ is the stress in pascals (Pa) and x is the distance from the fixed end in metres. Calculate the rate of change of stress of at 6 meters.**

Sol: 1. Stress function: $S(x) = 400x - x^3$

2. First derivative (rate of change):

$$\frac{dS}{dx} = 400 - 3x^2$$

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3. Evaluate at $x = 6$:

$$\frac{dS}{dx}|_{x=6} = 400 - 3(6)^2$$

$$\frac{dS}{dx}|_{x=6} = 400 - 108$$

$$\frac{dS}{dx}|_{x=6} = 292 \ \ Pa/m$$

Q.8. **The cost $C(r)$ to construct a cylindrical tank depends on the radius of the base, and is given by $C(r) = 8000\pi r^2 + \frac{150000}{r}$, where the first term represents the cost of the base and the second term represents the cost of the walls. Determine the rate of change of the cost at $r = 4$ metres.**

Sol: Find the derivative of $C(r)$

$$C(r) = 8000\pi r^2 + \frac{150000}{r}$$

$$C'(r) = 16000\pi r - \frac{150000}{r^2}$$

$$C'(4) = 16000\pi(4) - \frac{150000}{4^2} = 64000\pi - \frac{150000}{16}$$

$$C'(4) = 64000\pi - 9375 = 6400(3.1416) - 9375 = 191687.4 \text{ unit m.}$$

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